UM-2 solution

An accurate approximate solution of the coupled field equations, with Kohn-Sham theory for the dirac16complex field coupled to gravity and to the Standard-Model fields

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Solution, version 1, 2026-10-17

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1 How to read this document

This document presents the approximate solution of the coupled field equations of UM-2 and the verification of each step. It is self-contained: every equation it uses is written here. Labels: AUTHOR, RECORD, STANDARD,

ASSUMED, INVENTED, DERIVED (computed and verified; the verification is stated), ESTIMATE, OPEN. Units $\hbar = c = 1$; in the numerical work $H = 1$ and $m = H$ unless a physical unit is given. The coordinates are the author's $x_0, x_1, \ldots, x_7$ (x_0 the hidden space direction, x_1, x_2, x_3 space, x_4 the time, x_5, x_6, x_7 the extra times); W^+ carries the electric charge $+e$ and W^- the charge $-e$.

Symbols with more than one use: k is the block variable $1/f^2$ (k_*, δk, $G(k)$), the Lovelock order as the index of α_k, $E_{(k)}$, $\mathcal{L}_{(k)}$ (written p where the block variable appears in the same formula), the momentum of a mode along x_1, x_2, x_3 in h_j, $\vec{k}$, Δk and $\varepsilon = e^{-a_4}|k|$, and the comoving Fourier wavenumber in the perturbation equations of step 5. $\Lambda = -10H^2$ is the eight-dimensional cosmological constant and Λ_4 the four-dimensional one; in the x_0 eigenvalue problems of step 2, $\Lambda = (m\,fL)^2$ is a dimensionless eigenvalue; $\Lambda_\star$ is the scale of the effective interactions. n is a number density, except as a level number ($n\pi$) and in the lattice vector $\vec{n} \in \mathbb{Z}^3$ of a momentum shell; N is a particle number. The subscript t (p_t, v_t) refers to the extra times x_5, x_6, x_7; $\xi = y/L$ is the scaled hidden coordinate; primes denote $d/d\xi$ in step 2 and $d/d\tau$ (conformal time) in step 5. f is the block scale and f_o an occupation number.

2 The equations being solved

The action of UM-2 is

$$S = \frac{1}{2\kappa}\int d^8x\sqrt{|g|}\Big(\sum_{k=1}^{3}\alpha_k\mathcal{L}_{(k)} - 2\Lambda\Big) - \int d^8x\sqrt{|g|}\,\rho_b(f) + S_\Psi + S_{gauge} + S_f + S_{tip} + S_{I9} + S_{I11} + S_{I12},$$

with the author's $\alpha_k = w_k H^{2-2k}$, $w = (1, \frac{1}{32}, \frac{1}{1152})$, $\Lambda = -10H^2$; the stabiliser $\kappa\rho_b = \kappa\rho_b^*(f_*/f)^8$; the dirac16complex Lagrangian $\sqrt{|g|}[\frac{1}{2}(\bar\Psi\gamma^M D_M\Psi - (D_M\bar\Psi)\gamma^M\Psi) - m\bar\Psi\Psi - \frac{\lambda}{2}(\bar\Psi\Psi)^2]$ with the canonical spin connection; the eight-dimensional $SU(3)_c \times SU(2)_L \times U(1)_Y$ gauge fields and one generation of fermions; the tip sigma field $\Sigma \in SU(2)$ (no Higgs particle); and the portals

$$S_{I9} = \int_{y=-L} d^7x\sqrt{|g_{tip}|}\,(c_Z + c'_Z\varsigma)(i\bar\Psi\gamma^\mu P_+\Psi)\mathcal{Z}_\mu, \qquad S_{I11} = \int d^8x\sqrt{|g|}\,\frac{c_V}{\Lambda_\star^6}(i\bar\Psi\gamma^M\Psi)\sum_\psi q_\psi(i\bar\psi\gamma_M\psi),$$

$$S_{I12} = -\int d^8x\sqrt{|g|}\,\frac{\bar\Psi\Psi}{\Lambda_\star^3}\big(c_G G^\alpha_{MN}G^{\alpha MN} + c_W W^a_{MN}W^{aMN} + c_B B_{MN}B^{MN}\big),$$

$\mathcal{Z}_\mu = \frac{iv_\Sigma^2}{2}\mathrm{tr}(\Sigma^\dagger D_\mu\Sigma\tau^3) \to \frac{v_\Sigma^2}{2}\sqrt{\hat g^2 + \hat g'^2}Z_\mu$, $P_+ = (1+\Gamma_4)/2$. The field equations are

$$\sum_k \alpha_k {E_{(k)}}^M{}_N + \Lambda\delta^M_N = \kappa T^M{}_N, \qquad \gamma^M D_M\Psi = \Big[m + \lambda\bar\Psi\Psi + \frac{\mathcal{F}}{\Lambda_\star^3}\Big]\Psi - i\frac{c_V}{\Lambda_\star^6}\mathcal{J}_M\gamma^M\Psi - i\delta_{tip}(c_Z + c'_Z\varsigma)\mathcal{Z}_\mu\gamma^\mu P_+\Psi,$$

together with the Yang-Mills equations with fermion, tip and portal currents, the Dirac equations of the fermions with the tip Yukawa terms, and the boundary conditions at the fold and the tip.

3 The approximation scheme

The solution uses the small parameters of the problem:

- $\epsilon \sim 10^{-5}$: the metric is $g = \bar g + \epsilon h^{(1)} + \ldots$ (order 0: the background; order 1: the CMB, CNB, BAO);
- $(A/H)^2 \ll 1$ after the primordial era: the matter sources are tiny compared with the stabiliser (today 10^{-92}), so the block responds adiabatically and linearly;
- the Standard-Model masses are far below the gap $3H/(2f)$ of the x_0 operators: four-dimensional effective fields;
- dirac16complex is treated by Kohn-Sham density-functional theory on the static block, with the portal couplings as external mean fields (the Standard-Model fields enter through their densities and condensates).

The hierarchy: (1) the background and the primordial era; (2) the x_0 spectra, i.e. all masses; (3) the Kohn-Sham states of Ψ coupled to gravity and to the Standard-Model fields, and their back-reaction on the block; (4) the electroweak sector, N_{eff} and the CNB; (5) the linear cosmology; (6) the dark-matter relic.

4 Step 1: the background

4.1 The static block

The order-0 metric is

$$ds^2 = e^{2a_4}d\vec{x}^{\,2} - dx_4^2 + f^2\big[dy^2 - e^{2Hy}d\vec{x}_t^{\,2}\big], \qquad y = \frac{\ln|\sin 6Hx_0|}{6H}, \quad f = e^{-a_4}b_4^{1/2},$$

periodic in x_0 with period $\pi/(6H)$ (two halves joined at the fold $y=0$, a regular tip at $y=-L$, $\eta = HL = 3$). With $k = 1/f^2$, $A = \dot{a}_4$, $h = \dot{f}/f$, a static block ($h=0$, $A=0$) with the stabiliser ($p_3 = -\rho_b$, $p_0 = p_t = \rho_b$) requires (DERIVED)

$$\kappa\rho_b = 10H^2 - 6H^2k + \tfrac{3}{8}H^2k^2 = 3H^2k - 10H^2 \implies k_* = 12 + \tfrac{4}{3}\sqrt{51} = 21.5219,$$

$$f_* = 0.215556, \qquad G(k_*) = 1 - 24w_2k_* + 72w_3k_*^2 = \tfrac{20}{3} + \sqrt{51} = 13.8081 > 0,$$

$$\kappa\rho_b^* = (26 + 4\sqrt{51})H^2 = 54.566H^2.$$

$G(k_*) > 0$: four-dimensional gravity is attractive with the author's weights and Λ unchanged. (Without the stabiliser the only static block with $A=0$ is $k_0 = 8$ with $G = -1$, repulsive.) The linearised system has the eigenvalues $0, \pm i\omega_r$,

$$\omega_r^2 = \Big(\frac{31552}{2249} - \frac{2416\sqrt{51}}{6747}\Big)H^2, \qquad \omega_r = 3.38705H,$$

so the block is stable; the radion mass is $m_r = 3.387H = 48.38$ TeV. The whole stabiliser family $p_0 = \nu\rho_b$ ($\nu = 1/4$ to 3) gives stable blocks with $\omega_r = 3.33H$ to $4.10H$ (DERIVED).

4.2 The primordial era: the deflating extra times

The full nonlinear equations (all three components, the constraint preserved to 6×10^{-10}) were integrated from an expanding 3-space and a block larger than the stabilised one ($k_0 < k_*$) that deflates at $h_0 = -H/2, -H/3, -H/10$, for k_0 from 1.5 to 20 (DERIVED):

start k_0	$h_0 = -H/10$	$h_0 = -H/3$	$h_0 = -H/2$
1.5	singular at $x_4 = 1.44/H$	singular at $1.26/H$	singular at $1.14/H$
2.3	singular at $0.33/H$	singular at $0.31/H$	no expanding start
2.7	$k(60/H) = 21.415$	21.435	21.451
3	21.412	21.434	21.455
6	21.405	21.453	21.500
10	21.446	21.541	21.584
20	21.634	21.640	21.638

Every start above the separatrix $k_c = 2.5301, 2.5991, 2.6983$ (for $h_0 = -H/10, -H/3, -H/2$; just above the unstable static block $k_- = 2.4781$) ends at the stable block: the extra times x_5, x_6, x_7 and the hidden direction x_0 deflate at rates up to $|h| = 2.63H$ (in the stored runs $k_0 = 3, 5, 10$), by up to $f(k_c)/f_* = 2.917$, reach k_* within $x_4 < 1/H$, and then oscillate about it with the radion frequency ($3.3868H$ measured, $3.3870H$ linear), while 3-space keeps expanding; the oscillation energy redshifts like dust ($A\,x_4 \to 0.667$ to 0.670; 2/3 for dust). The extra times therefore deflate exponentially only in a short primordial era and are then frozen: the masses, all of which depend on f, become constant.

4.3 The response of the block to matter, and four-dimensional gravity

To first order about the static block with block-averaged matter sources ($\bar{p} = (p_0 + 3p_t)/4$; A^2 of first order, $3G(k_*)A^2 = \kappa\delta\rho$) (DERIVED):

$$\delta\ddot{k} + \omega_r^2\,\delta k = -2k_*\big(s_{h3}\,\kappa\delta p_3 + s_{hp}\,\kappa\delta\bar{p} + q_h\,\kappa\delta\rho\big), \qquad s_{h3} = 0.028666, \quad s_{hp} = 0.037320, \quad q_h = -0.009555;$$

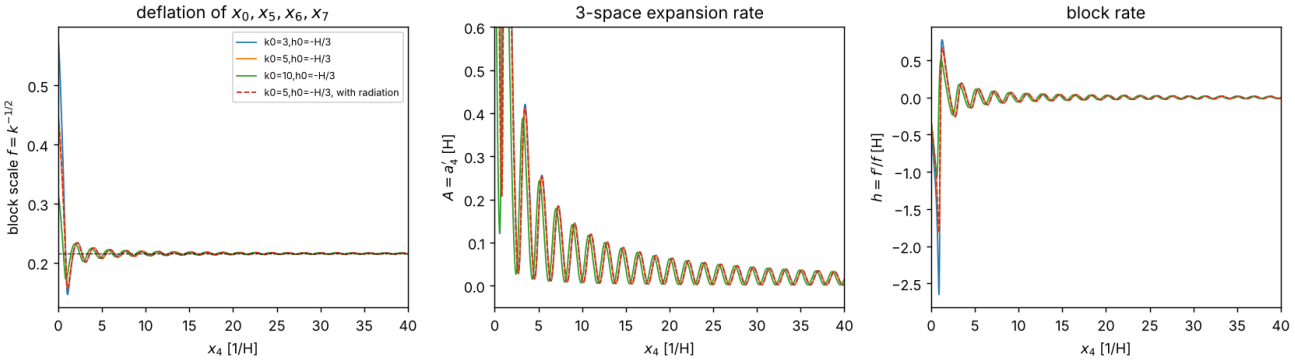


Figure 1: The primordial era: the block scale f, the 3-space expansion rate A and the block rate h for starts $k_0 = 3, 5, 10$ at $h_0 = -H/3$ (one with radiation), from the full nonlinear Einstein-Lovelock equations with the author's weights.

the adiabatic shift is $\delta k = (-0.107556\,\kappa\delta p_3 - 0.140028\,\kappa\delta\bar{p} + 0.035852\,\kappa\delta\rho)/H^2$, and 3-space obeys $\delta\dot{A} = -\kappa(\delta\rho + \delta p_3)/(2G(k_*))$. Together with the constraint $3G(k_*)A^2 = \kappa\rho$ and the dimensional reduction $(\sum_p \alpha_p \mathcal{L}_{(p)} - 2\Lambda - 2\kappa\rho_b = G(k)R^{(4)} + \text{topological} - 2G(k)\Lambda_4$, p the Lovelock order), the four-dimensional gravity of UM-2 is general relativity,

$$G^{(4)}_{\mu\nu} + \Lambda_4 g_{\mu\nu} = 8\pi G_N T^{(4)}_{\mu\nu}, \qquad 8\pi G_N = \frac{\kappa}{G(k_*)V_\perp}, \qquad V_\perp = \frac{2f_*^4 v_t(1 - e^{-3\eta})}{3H},$$

with $\Lambda_4 = [6H^2k(1 - 2w_2k) + \Lambda + \kappa\rho_b]/G(k) = 0$ at the static point; the observed dark energy needs $\Lambda_4/H^2 \sim (H_0/H)^2 = 1.0 \times 10^{-92}$ (tuned; OPEN).

5 Step 2: the masses from the x_0 dependence of the wave functions

Every four-dimensional mass is an eigenvalue of the x_0 operator of its field on the static block; the eigenfunctions depend on x_0 through $y(6Hx_0)$ and are periodic with period $\pi/(6H)$ (the author's requirement). With $\xi = y/L$:

5.1 Gauge bosons and the graviton

$$-e^{-3\eta\xi}\left(e^{3\eta\xi}u'\right)' = \Lambda u, \qquad \Lambda = (m\,fL)^2, \qquad u'(0) = 0 \text{ (even) or } u(0) = 0 \text{ (odd)}, \qquad u'(-1) = \beta u(-1),$$

$$\text{even: } \left[q^2 + \frac{3\eta}{2}\left(\beta + \frac{3\eta}{2}\right)\right]\sin q - \beta q\cos q = 0, \qquad \text{odd: } q\cos q + \left(\beta + \frac{3\eta}{2}\right)\sin q = 0, \qquad q^2 = \Lambda - \frac{9\eta^2}{4}.$$

mode ($\eta = 3$)	$\Lambda = (m\,fL)^2$	status
photon, gluons, graviton ($\beta = 0$, flat profile)	0 exactly	DERIVED
first even Kaluza-Klein level	$9\eta^2/4 + \pi^2 = 30.1196$	DERIVED (finite elements 1.9×10^{-10})
first odd level	27.0890 ($q\cot q = -9/2$)	DERIVED (finite elements 4.5×10^{-10})
gap	$9\eta^2/4 = 20.25$	DERIVED
W^+, W^- ($\beta_W = 0.011932$)	1.32374×10^{-5}	DERIVED (fit, finite elements 3×10^{-14})
Z ($\beta_Z = \beta_W/\cos^2\theta_W = 0.015364$, on-shell $\cos^2\theta_W = m_W^2/m_Z^2 = 0.77666$)	1.70374×10^{-5}	DERIVED
Dirichlet bound ($\beta \to \infty$)	0.010016: $m_W \le 0.01824\,m_{KK}$	DERIVED

β_W is fixed by the measured $\rho_0 - 1 = 0.00038$: the bending of the tip-localised profiles moves the model's mass ratio $m_W^2/m_Z^2 = \Lambda_W/\Lambda_Z$ away from the on-shell $\cos^2\theta_W$ of the measured masses, and the electroweak parameter $\rho = \Lambda_W/(\Lambda_Z \cos^2\theta_W)$ exceeds 1 ($\rho - 1 = 3.2\times10^{-8}$ at $\beta_W = 10^{-6}$: custodial symmetry at leading order; 6.4×10^{-5} at $\beta_W = 0.002$, 3.2×10^{-3} at 0.1). With $m_W = 80.3625$ GeV this gives the scales (DERIVED)

quantity	value
Kaluza-Klein mass $m_{KK} = m_W\sqrt{\Lambda_1/\Lambda_W}$	121.22 TeV
$H = m_{KK} f_* \eta/\sqrt{\Lambda_1}$	14.283 TeV
proper half cell fL; period of x_0, $\pi/(6H)$	8.934×10^{-21} m; 7.234×10^{-21} m
gap $3H/(2f_*)$	99.40 TeV
radion mass $3.387H$	48.38 TeV
model $m_Z = m_W\sqrt{\Lambda_Z/\Lambda_W}$	91.171 GeV (measured 91.188)

5.2 Fermions

$$h_j = j\Big[-\frac{i}{f}\sigma_1\partial_y + \mu\,\sigma_2 + e^{-a_4}k\,\sigma_3\Big],$$

$$D(e) = e^2 s_L s_R - \kappa_f^2 (C_L - c_L s_L)(C_R + c_R s_R) = 0, \qquad s = \frac{\sinh w}{w}, \quad C = \cosh w, \quad w = \sqrt{c^2 - e^2},$$

with k the momentum of the mode along x_1 (not the block variable) and $c = f\mu L$; the first-order solution $e = \kappa_f \zeta(c_L)\zeta(-c_R)$, $\zeta(c)^2 = c\,e^{-c}/\sinh c$ ($c_L > 0$, $c_R < 0$: both zero modes at the fold), agrees with the exact root to 1.2×10^{-8}, and direct shooting of the coupled Dirac system agrees with the determinant to 3.5×10^{-10}. With $\kappa_f = 1$ and $c_L = -c_R = c$ (DERIVED):

fermion	mass	c	μ/H	proper width $fL/(2c)$	$e = m\,fL$
ν	0.05 eV	18.6709	28.873	2.39×10^{-22} m	2.264×10^{-15}
e	0.511 MeV	10.3038	15.934	4.34×10^{-22} m	2.313×10^{-8}
u	2.16 MeV	9.5448	14.760	4.68×10^{-22} m	9.779×10^{-8}
d	4.70 MeV	9.1341	14.125	4.89×10^{-22} m	2.128×10^{-7}

5.3 dirac16complex

With $M = m$, fold condition $\chi_2(0) = 0$ (even) or $\chi_1(0) = 0$ (odd), tip $\chi_2(-L) = 0$ and $c_\psi = f_* mL = 0.646667$, the exact $k = 0$ spectrum is: the fold zero mode ($\varepsilon = 0$, all eight blocks); even bulk levels $\sqrt{m^2 + (n\pi/(fL))^2}$, the lowest $4.959981\,m$; odd levels $\sqrt{m^2 + p^2}$ with $\tan(p\,fL) = -p/m$, the lowest $3.102204\,m$ ($p = 2.93661\,m$, $p\,fL = 1.89901$); and the fold band $\varepsilon = e^{-a_4}|k|$ (massless, $\varepsilon/|k| = 1$ to 6×10^{-14}) (DERIVED).

6 Step 3: Kohn-Sham states of dirac16complex coupled to all fields

6.1 The Kohn-Sham equations

$$h_j\chi_o = \varepsilon_o\chi_o, \qquad M_{eff} = m + \frac{15}{16}\lambda S + \frac{\langle\mathcal{F}\rangle}{\Lambda_\star^3}, \qquad v = -\frac{\lambda}{16}n + \frac{c_V}{\Lambda_\star^6}\sum_\psi q_\psi n_\psi + v_{I9},$$

$$n = \sum wgf_o n_o, \quad S = \sum wgf_o s_o, \quad n_o = P\chi_o^\dagger\chi_o, \quad s_o = P\chi_o^\dagger j\sigma_2\chi_o, \quad t_o = P\chi_o^\dagger j\sigma_3\chi_o, \quad P = \frac{e^{-3Hy}}{\mathrm{Vol}_7},$$

$$E_{KS} = \sum gf_o\varepsilon_o - 2\mathrm{Vol}_7\int e^{3Hy}e_{int}dy, \qquad e_{int} = \frac{15}{32}\lambda S^2 - \frac{1}{32}\lambda n^2,$$

$$\rho = \sum wgf_o\varepsilon_o n_o - e_{int}, \qquad p_3 = \sum wgf_o\tfrac{e^{-a_4|k|}}{3}t_o + e_{int}, \qquad p_t = e_{int},$$

$$p_0 = \sum wgf_o[(\varepsilon_o - v)n_o - M_{eff}s_o - e^{-a_4}|k|t_o] + e_{int},$$

with $w = 1/2$ per fold parity, $g = 4r_3(|\vec{n}|^2)$ for the momentum shell $\vec{k} = \Delta k\,\vec{n}$ ($\vec{n} \in \mathbb{Z}^3$, $r_3(|\vec{n}|^2)$ the number of representations of $|\vec{n}|^2$ as a sum of three squares, $\Delta k = 0.25$; here k is the momentum, not the block variable), occupations f_o, and $\mathrm{Vol}_7 = (2\pi/\Delta k)^3 v_t f_*^4 = 34.27$ ($v_t = 1$, $a_4 = 0$; the proper seven-volume of a cell is $2\,\mathrm{Vol}_7(1 - e^{-3\eta})/(3H)$). The orbitals are expanded in the exact free eigenfunctions of each sector (60, 120, 240 per sector), iterated to self-consistency (tolerance 10^{-13}, safeguarded Anderson mixing), and extrapolated in the basis size (Richardson).

6.2 Validation

The independent solver reproduces the record's two reference states (at $f = 1$, $a_4 = 0$): $N = 8$, $\lambda = 157.7$: $E_{KS} = -0.037786682923$ (reference -0.037786682923), HOMO and LUMO to 10^{-14}, integrated $p_0 = -0.08860537545$ (reference -0.08860537548); $N = 592$, $\lambda = 2.131$: $E_{KS} = 358.30104644787$ (identical), integrated $p_0 = -6.7691415097$ (reference -6.7691415108) (DERIVED).

6.3 Ground states on the stabilised block

Two families of states (the eight $k = 0$ fold zero modes, $N = 8$; a fold-band gas, $N = 592$) with $\lambda = 0$, the benchmark $\pm\lambda_B$ ($\lambda_B = 7 \times 10^{-5} H^{-6}$: mean fields below 5 % of m) and the record's coupling scaled to the block (stress test, $\lambda = 157.7 f_*^4 = 0.3405$ and $2.131 f_*^4 = 0.00460$) (DERIVED; 120 functions per sector, Richardson errors below 1.5×10^{-9}):

state	E_{KS}	E_{int}	HOMO	LUMO	iterations
$N = 8$, $\lambda = 0$	0	0	0	0.25	1
$N = 8$, $+\lambda_B$	-2.053×10^{-4}	-2.053×10^{-4}	-5.13×10^{-5}	0.24995	14
$N = 8$, $-\lambda_B$	2.053×10^{-4}	2.053×10^{-4}	5.13×10^{-5}	0.25005	14
$N = 8$, stress	-0.70123	-0.42170	-0.14037	0.10859	61
$N = 592$, $\lambda = 0$	361.11614	0	0.79057	0.82916	1
$N = 592$, $+\lambda_B$	359.99060	-1.12646	0.78676	0.82535	14
$N = 592$, $-\lambda_B$	362.23906	1.12121	0.79436	0.83295	13
$N = 592$, stress	307.67139	-32.2217	0.64506	0.68347	56

state	$\int p_3$	$\int p_t$	$\int p_0$	max $\lvert\delta M\rvert$	max $\lvert v\rvert$
$N = 8$, $+\lambda_B$	-2.05×10^{-4}	-2.05×10^{-4}	-4.447×10^{-4}	2.4×10^{-7}	6.7×10^{-4}
$N = 8$, $-\lambda_B$	2.05×10^{-4}	2.05×10^{-4}	4.447×10^{-4}	2.4×10^{-7}	6.7×10^{-4}
$N = 8$, stress	-0.4217	-0.4217	-1.30076	1.98	1.83
$N = 592$, $\lambda = 0$	120.372	0	8.6×10^{-12}	0	0
$N = 592$, $+\lambda_B$	119.245	-1.1265	-2.43688	1.3×10^{-3}	0.050
$N = 592$, $-\lambda_B$	121.493	1.1212	2.43174	1.3×10^{-3}	0.050
$N = 592$, stress	87.614	-32.222	-97.8188	2.09	1.88

(The integrals are $2\mathrm{Vol}_7 \int e^{3Hy}(\cdot)\,dy$; the $N = 8$, $\lambda = 0$ state has all of them zero.)

Checks (DERIVED): $E_{KS} = 2\mathrm{Vol}_7 \int e^{3Hy}\rho\,dy$ to 4×10^{-15}; the y-conservation law $[e^{3Hy}p_0]_{-L}^{0} = 3H \int e^{3Hy}p_t dy$ to 9×10^{-8} (Richardson; the raw residual falls like n_b^{-3}); observed basis orders 4.95 (energy) and 2.97 (hidden pressure); the free gas has $w = \int p_3 / \int \rho = 1/3$ exactly (four-dimensional radiation). The self-interaction shifts the fold-band levels almost rigidly (the spacing between $k = 0$ and the first shell is 0.25 free, 0.25000 at λ_B, 0.2490 in the $N = 8$ stress state). A scalar mean field cannot gap the fold band at $k = 0$: $\chi = (e^{f\int M_{eff}dy}, 0)$ solves $j[-\frac{i}{f}\sigma_1\partial_y + M_{eff}\sigma_2]\chi = 0$ with both boundary conditions for every $M_{eff}(y)$.

6.4 Dark-matter quanta

Eight quanta in the lowest odd level at $k = 0$ on the $N = 8$ ground state add exactly $8\varepsilon_{DM} = 24.817633063$ (computed 24.817633063) at $\lambda = 0$ and 24.81833 at $+\lambda_B$; they are cold, $w = p_3/\rho = 0$ (2.8×10^{-5} with λ_B); their hidden pressure is $p_0/\rho = 0.772034343$, equal to $-d\ln\varepsilon_{DM}/d\ln f = 0.772034343$ obtained independently from the x_0 eigenvalue (DERIVED): the dark-matter mass is an x_0 eigenvalue, and it pushes on the block.

6.5 The Standard-Model fields acting on dirac16complex

The portal mean fields were applied with amplitude e and shapes given by the sources (the electron and up-quark zero-mode densities times P for I11; a uniform gluon condensate for I12; a tip-localised Z charge on the upper component for I9). Linear responses of the lowest odd level (DERIVED; equal to the Hellmann-Feynman expectation $d\varepsilon/de = \langle\chi|\partial h/\partial e|\chi\rangle$ to 10^{-8}):

portal mean field	$d\varepsilon_{odd}/de$	change of the fold-band gap
I11, electron density (v)	0.507981	1.2×10^{-10}
I11, up-quark density (v)	0.548555	1.6×10^{-10}
I12, gluon condensate (uniform M)	0.707196	2.9×10^{-11}
I9, tip Z charge (upper component): Ψ_A odd level; Ψ_B (the dark matter)	0.259489; exactly 0	7.9×10^{-11}

The fold band stays massless under every portal. Every orbital has $\chi_2(-L) = 0$, so the scalar density $\bar\Psi\Psi = 2s_2ab$ vanishes at the tip: no tip operator built on $\bar\Psi\Psi$ reaches dirac16complex, and I9 acts on the upper component only. For Ψ_B ($B = +1$) the tip projector $P_+ = (1 - B\gamma^{x_0})/2$ annihilates every orbital at the tip, so the Ψ_B dark matter has no I9 coupling at all (section 7). The I12 response defines the four-dimensional coefficient of the dark-matter operator: $G_i = \mathcal{O}\, c_i g_i^2/\Lambda_\star^3$ with $\mathcal{O} = 0.7072$.

6.6 Back-reaction on the block

The block-averaged hidden pressure $\bar p = (p_0 + 3p_t)/4$ of a Kohn-Sham state displaces the block by $\delta k = \kappa\bar p/(3H^2 - 4\kappa\rho_b^*/k_*) = -0.140028\,\kappa\bar p/H^2$ (stiffness $-7.1414H^2$), without changing the four-dimensional cosmological term at linear order ($\frac{d}{dk}[6H^2k(1 - 2w_2k) + \kappa\rho_b] = -6 \times 10^{-12}$ at k_*). Cell averages (units of the Kohn-Sham calculation, $H = 1$): $\bar p = -1.16 \times 10^{-5}$ ($N = 8$, $+\lambda_B$), -0.0636 ($N = 592$, $+\lambda_B$), ≈ 0 (free gas); the shift is negligible for any realistic density (relative size $(A/H)^2$). The difference $p_0 - p_t$ (-0.0574 for $N = 592$, $+\lambda_B$) drives a block-shape mode outside the order-0 metric (OPEN).

7 Step 4: the electroweak sector, N_{eff} and the CNB

Charges and couplings (DERIVED): $Q = T_3 + Y$ gives $(u,d)_L = (2/3, -1/3)$, $(\nu, e)_L = (0, -1)$, $u_R = 2/3$, $d_R = -1/3$, $e_R = -1$, $\nu_R = 0$; $[T_3, T_\pm] = \pm T_\pm$ with $Y(W) = 0$: $W^+ = (W^1 - iW^2)/\sqrt{2}$ **carries** $+e$, $W^- = (W^1 + iW^2)/\sqrt{2}$ **carries** $-e$, and W^- is the antiparticle of W^+; the photon couples with $e = gg'/\sqrt{g^2 + g'^2}$ times Q, the Z with $\sqrt{g^2 + g'^2}(T_3 - Q\sin^2\theta_W)$; the one generation is anomaly free ($SU(3)^2U(1)$, $SU(2)^2U(1)$, $U(1)^3$, gravitational, $SU(3)^3$ all vanish; four doublets, no Witten anomaly).

The two halves of dirac16complex. B commutes with the four-dimensional Lorentz generators and with C, so $\Psi = \Psi_A + \Psi_B$ ($B = -1$, $B = +1$) splits into Lorentz-invariant halves and $\bar\Psi\Psi = \bar\Psi_A\Psi_A + \bar\Psi_B\Psi_B$; the identity $\Gamma_4 = -B\gamma^{x_0}$ (DERIVED, 16×16) gives $P_+ = (1 - B\gamma^{x_0})/2$, which equals $(1 + \gamma^{x_0})/2$ on Ψ_A and $(1 - \gamma^{x_0})/2$ on Ψ_B; the latter annihilates every orbital at the tip. Hence the tip Z charge densities (DERIVED): Ψ_A fold band $a_0(-L)^2 = 0.163$ per unit length in each of its four blocks, Ψ_B fold band 0; Ψ_A odd (heavy) states 0.257, Ψ_B odd (dark-matter) states 0, and every transition current $\bar\Psi_A\gamma^\mu P_+\Psi_B$ vanishes at the tip.

The Z invisible width. With one neutrino and the two Ψ_A Weyl species of Z-coupling r times the neutrino's,

$$N_\nu = \frac{\Gamma_{inv}}{\Gamma(Z \to \nu\bar\nu)} = 1 + 2r^2 = 2.9840 \pm 0.0082 \implies r^2 = 0.9920 \pm 0.0041, \qquad r = 0.99599.$$

N_{eff} and the CNB. The neutral-current rate scales as $r^2 G_F^2 T^5$ against $H \propto T^2/M_{Pl}$, so Ψ_A decouples at $T \approx 2.6\,\mathrm{MeV} \times r^{-2/3} = 2.61$ MeV, after the QCD and pion entropy releases; $T_{\Psi_A} = T_\nu$ for every $r^2 > 2.2 \times 10^{-3}$, and

$$N_{eff} = 3.044, \qquad T_\nu = \left(\frac{4}{11}\right)^{1/3} T_\gamma = 1.9454\ \mathrm{K}, \qquad n = \frac{3}{4}\frac{\zeta(3)}{\pi^2}\, 2T_\nu^3 = 112.0\ \mathrm{cm}^{-3}\ \text{per Weyl species.}$$

The cosmic neutrino background of UM-2: ν_e (0.05 eV, non-relativistic today) and two massless Ψ_A species, 336 cm^{-3} in all (DERIVED). The Ψ_B fold band has no tip coupling and no scalar density and, with $c_V = 0$, is never thermalised.

The Z width of one generation (tree level, $\Gamma(Z \to f\bar{f}) = N_c G_F m_Z^3 (g_V^2 + g_A^2)/(6\sqrt{2}\pi)$, $g_V = T_3 - 2Q\sin^2\theta_W$ with the $\overline{MS}$ value $\sin^2\theta_W = 0.23122$, $g_A = T_3$):

channel	ν_e	e	u	d	Ψ_A (2 Weyl)	total	measured
width (MeV)	165.9	83.4	285.4	367.9	329.1	1231.7	2495.5

The invisible width is reproduced through the fit of N_ν (tree level 495 MeV against the measured 499.0 ± 1.5 MeV); the total is not, because the order restricts the particle content to one generation (the μ, τ, s, c, b channels are missing): an inherent conflict of the specified content with data.

8 Step 5: the CMB, the CNB and the BAO

The four-dimensional effective theory is general relativity (step 1) with photons, baryons, ν_e, the two Ψ_A species, the Ψ_B dark matter and Λ_4. Its linear perturbations obey the standard Einstein-Boltzmann equations (Newtonian gauge, conformal time, $ds^2 = a^2[-(1+2\psi)d\tau^2 + (1-2\phi)d\vec{x}^{\,2}]$):

$$k^2\phi + 3\mathcal{H}(\phi' + \mathcal{H}\psi) = -4\pi G_N a^2 \delta\rho, \qquad k^2(\phi' + \mathcal{H}\psi) = 4\pi G_N a^2(\bar{\rho} + \bar{P})\theta,$$

$$\phi'' + \mathcal{H}(\psi' + 2\phi') + (2\mathcal{H}' + \mathcal{H}^2)\psi + \frac{k^2}{3}(\phi - \psi) = 4\pi G_N a^2 \delta P, \qquad k^2(\phi - \psi) = 12\pi G_N a^2(\bar{\rho} + \bar{P})\sigma,$$

with the cold dark matter $\delta_c' = -\theta_c + 3\phi'$, $\theta_c' = -\mathcal{H}\theta_c + k^2\psi$, the baryon-photon fluid with Thomson scattering, the massless hierarchy for Ψ_A and the massive hierarchy for ν_e. They were solved with CAMB 2.0.5 at the Planck 2018 values $\omega_b = 0.02237$, $\omega_c = 0.1200$, $100\theta_{MC} = 1.04092$, $\tau_{reio} = 0.0544$ (the optical depth to reionisation), $\ln(10^{10}A_s) = 3.044$, $n_s = 0.9649$ (θ held fixed, so H_0 is an output; H_0 in km/s/Mpc). The models are Planck 2018 ΛCDM (three neutrinos, $\sum m_\nu = 0.06$ eV), UM-2 (ν_e of 0.05 eV and two massless Ψ_A species) and the UM-2 variant with the Ψ_B fold band thermalised above the QCD crossover (DERIVED):

model	H_0	Ω_m	N_{eff}	Y_{He}	age (Gyr)	z_*	r_{drag} (Mpc)	BAO χ^2 (13 points)
ΛCDM	67.34	0.3154	3.044	0.24566	13.799	1089.91	147.103	32.55
UM-2	67.43	0.3143	3.044	0.24566	13.793	1089.91	147.105	30.31
UM-2, Ψ_B hot	70.62	0.2865	3.318	0.24925	13.515	1090.17	145.750	15.97

The UM-2 TT and EE spectra agree with Planck ΛCDM to 2.2×10^{-4} and 3.7×10^{-4} ($30 \le \ell \le 2500$), at most 0.011 of the cosmic variance $\sqrt{2/(2\ell+1)}$ of a single multipole: UM-2 reproduces the CMB of ΛCDM. If the Ψ_B fold band were thermalised and decoupled just above the QCD crossover, it would add $\Delta N_{eff} = 2(10.75/g_s)^{4/3} = 0.274$, with $g_s = 47.75$ counting the plasma it leaves (photons, gluons, u, d, e, ν_e and the two Ψ_A species) and 10.75 the same count at neutrino decoupling (0.218 for decoupling above the electroweak scale, $g_s = 56.75$). At fixed Planck

parameters this variant changes TT by up to 4.2 % for $30 \le \ell \le 2500$ (4.9 % at $\ell = 2$, where the cosmic variance of one multipole is 63 %; 2.1 cosmic-variance units of a single multipole; the full-sky cosmic-variance $\chi^2 = \sum_b (\bar{\delta}_b/\sigma_b)^2$ over bins b of $\Delta\ell = 50$, with $\bar{\delta}_b$ the mean relative deviation and $\sigma_b = \sqrt{2/\sum_{\ell\in b}(2\ell+1)}$, is 2314); it is not a fit to the CMB, and a joint CMB and BAO fit of it is OPEN (it is not excluded by this computation). The CNB enters through $N_{eff} = 3.044$ and the ν_e mass.

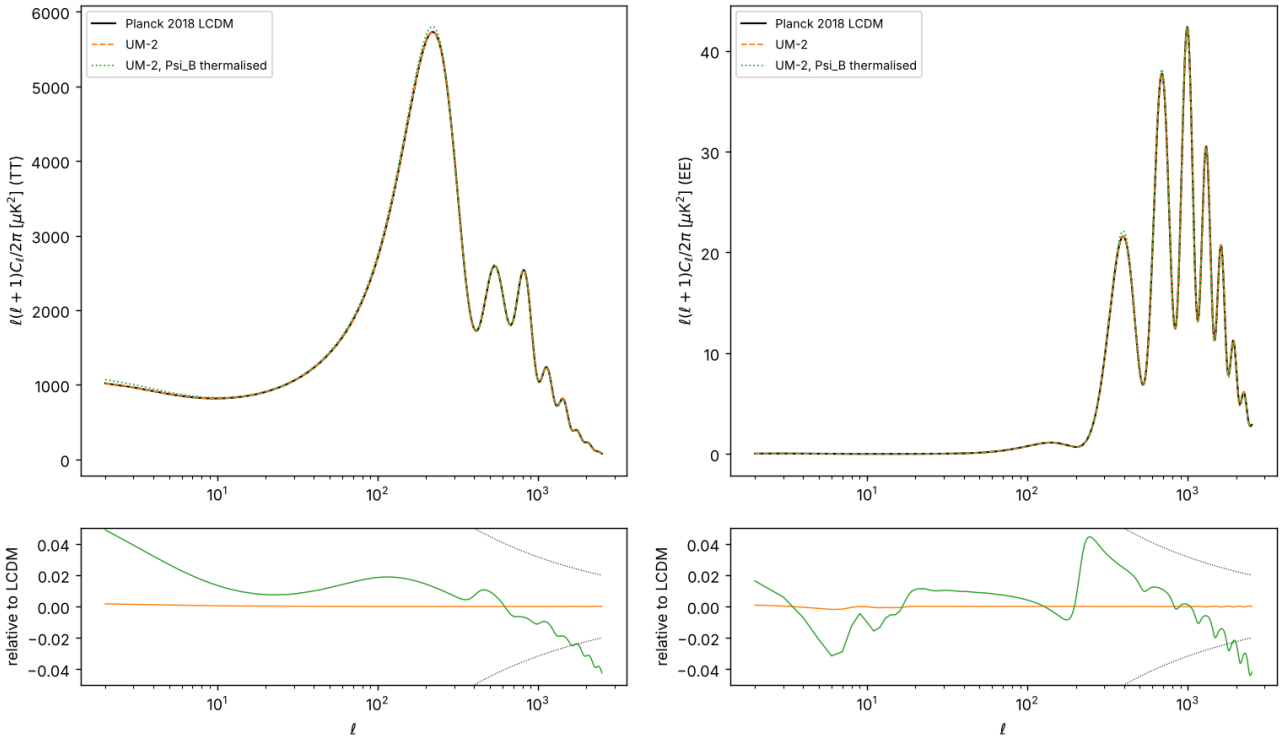


Figure 2: CMB temperature (left) and E-mode (right) spectra of UM-2, of Planck 2018 ΛCDM and of the variant with a thermalised Ψ_B fold band; lower panels: relative difference to ΛCDM with the cosmic variance of one multipole (dotted).

9 Step 6: the dark-matter relic and its detection

The lightest odd state of Ψ_B is a Dirac fermion of mass $M_{DM} = 3.102204\, m = 44.31$ TeV ($m = H$), cold, neutral, with no tree-level Z coupling. Through I12 it annihilates in the p-wave into gauge-boson pairs: for $\mathcal{L} = G\,\bar{\chi}\chi F_{\mu\nu}F^{\mu\nu}$ per abelian channel, $\sigma v = G^2 s(s-4M^2)/(4\pi) \to G^2 M^4 v^2/\pi$, in total $\sigma v = (M^4 v^2/\pi)(8G_g^2 + 3G_W^2 + G_B^2)$ (the polarisation sum $\sum|(k_1\cdot k_2)(e_1\cdot e_2) - (k_1\cdot e_2)(k_2\cdot e_1)|^2 = s^2/2$ and the spin trace $\mathrm{Tr}[(\not{p}_1 + M)(\not{p}_2 - M)] = 2(s - 4M^2)$ checked numerically). Freeze-out (Dirac fermion, $g_* = 56.75$: the gauge bosons without a Higgs, e, ν_L, u, d and the two Ψ_A species; ν_R never thermalised):

$$x_f = \ln\left[0.038\, c(c+2)\, g\, M_{Pl} M \langle\sigma v\rangle/\sqrt{g_* x_f}\right] = 30.31, \qquad \Omega h^2 = \frac{2 \times 1.07 \times 10^9\,\mathrm{GeV}^{-1}\, x_f}{\sqrt{g_*}\, M_{Pl}\, 3b/x_f} = 0.120$$

($\langle\sigma v\rangle = 6b/x$, $c = 1/2$, $g = 2$, $M_{Pl} = G_N^{-1/2} = 1.22 \times 10^{19}$ GeV, standard radiation-dominated freeze-out) requires $\langle\sigma v\rangle(x_f) = 1.37 \times 10^{-25}$ cm^3/s, below the s-wave thermal unitarity estimate $(4\pi/M^2)\sqrt{x_f/\pi} = 2.3 \times 10^{-25}$ cm^3/s (an estimate made for an s-wave process, applied here to a p-wave one), i.e.

$$(8G_g^2 + 3G_W^2 + G_B^2)^{1/2} = (16.6\ \mathrm{TeV})^{-3}, \qquad G_i = \mathcal{O}\, c_i g_i^2/\Lambda_\star^3, \quad \mathcal{O} = 0.7072.$$

Direct detection (ESTIMATE): even if the whole annihilation went into gluons, the trace anomaly ($\langle N|G^\alpha G^\alpha|N\rangle = -\frac{8\pi}{9\alpha_s} m_N f_{TG}$) gives $\sigma_{SI} = \mu^2 f_N^2/\pi = 4.3 \times 10^{-53}$ cm^2, eight orders below the LZ limit extrapolated to 44 TeV (2.4×10^{-45} cm^2); the electroweak channels have no tree-level nucleon coupling. The relic value is conditional:

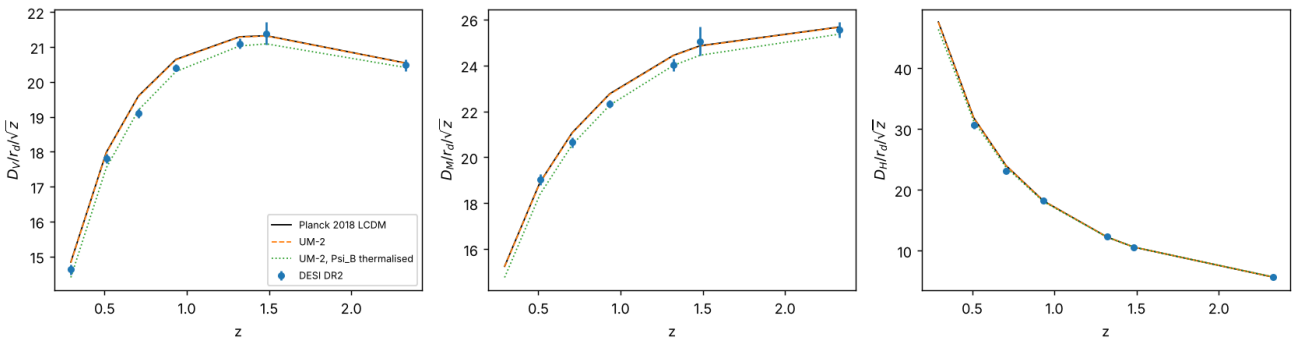


Figure 3: BAO distances divided by $\sqrt{z}$ against the DESI DR2 measurements; the χ^2 uses D_V for BGS and D_M, D_H with their correlation for the other tracers (the D_V points of the anisotropic tracers are shown for illustration).

if the radion oscillation left over from the deflation (which redshifts like dust) dominated the Universe until it decayed, it would dilute the dark matter after freeze-out (the moduli problem, below).

10 Accuracy of the solution

- Exact (symbolic): the background equations, the static block, $G(k_*)$, the radion frequency, the first-order block response, the dimensional reduction, the field reductions and the zero modes.
- Numerical, with measured errors: boson levels (mpmath, 30 digits; finite elements $\leq 5 \times 10^{-10}$); fermion masses (exact roots; first order 1.2×10^{-8}; shooting 3.5×10^{-10}); Kohn-Sham energies (Richardson errors below 1.5×10^{-9} relative; conservation 9×10^{-8}); primordial integrations (constraint below 6×10^{-10}); CMB (CAMB default accuracy, lensing included).
- Approximations: the static block with an assumed stabiliser; the good sector (no extra-time momenta); the Kohn-Sham mean-field treatment (local exchange of the contact interaction); portal couplings as external mean fields; tree level for the electroweak widths; the freeze-out formula; order ϵ for the cosmology. A fresh-clone rerun reproduced every stored number to 10^{-9} relative.

11 What the solution does not reproduce, and what is open

- **Conflicts with data, inherent in the order**: no Higgs boson (one is observed at 125 GeV), and without it the longitudinal W^+W^- scattering violates perturbative unitarity near $\sqrt{8\pi}\, v \approx 1.2$ TeV, far below the Kaluza-Klein modes; the total Z width of one generation, 1.23 GeV against 2.4955 GeV.
- **Open**: the origin of the stabiliser; the extra-time modes outside the good sector (tachyonic for momentum $q > fM$); the closed time-like curves implied by compact extra times; the radion left over from the deflation (ESTIMATE: a Planck-suppressed decay $\Gamma = C_r\, m_r^3/(4\pi \bar{M}_{Pl}^2)$, with the reduced Planck mass $\bar{M}_{Pl} = 2.435 \times 10^{18}$ GeV and an order-one coupling factor C_r, gives the lifetime $0.43\,\mathrm{s}/C_r$ and the reheating temperature $T_{RH} = (90/(\pi^2 g_*))^{1/4}\sqrt{\Gamma \bar{M}_{Pl}} = 1.85\sqrt{C_r}$ MeV, $g_* = 10.75$, below the 4.7 MeV that nucleosynthesis needs unless $C_r > 6.5$: a moduli problem); the block-shape mode driven by $p_0 - p_t$; the tuning of Λ_4 ($10^{-92} H^2$); the Kaluza-Klein graviton spectrum with the Lovelock corrections; the electroweak S parameter of the Higgsless tip; anomalies of the eight-dimensional bulk theory; the joint fit of the thermalised-Ψ_B variant.